

Nonlinear extensions of linear recurrence sequences

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Linear recurrence sequences (LRS)

$$a : \mathbb{N} \rightarrow \mathbb{Q}$$

For example $a_n = n^2$

$$a_0 = 0, \quad a_{n+1} = a_n + 2b_n + c_n$$

$$b_0 = 0, \quad b_{n+1} = b_n + c_n$$

$$c_0 = 1, \quad c_{n+1} = c_n$$

$$a_n = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}^n \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

In general $u_n = I^\top M^n F$

Can be also defined by $a_{n+3} = 3a_{n+2} - 3a_{n+1} + a_n$

In general $u_{n+k} = \sum_{i=0}^{k-1} \alpha_i u_{n+i}$

Many nice properties of LRS

- Unique, computable minimal degree

$$(d = 3 \text{ for } a_n = n^2)$$

- Closure properties, given LRS a_n, b_n :

$$c_n = a_n + b_n, \quad c_n = a_n \cdot b_n, \quad c_n = a_n * b_n = \sum_{i < n} a_i \cdot b_{n-i-1}$$

- Closed form

$$a_n = \sum_i p_i(n) \cdot \lambda_i^n$$

(for n big enough)

e.g. $F_n = \frac{1}{\sqrt{5}}\varphi^n + \frac{1}{\sqrt{5}}\psi^n$

Weighted automata over the rational field

- $\mathcal{A}$ is defined by $\Sigma = \{a, b, c\}$, $M_a, M_b, M_c \in \mathbb{Q}^{d \times d}$, $I, F \in \mathbb{Q}^d$

$$\mathcal{A} : \Sigma^* \rightarrow \mathbb{Q}$$

$$\mathcal{A}(w) = I^\top M_w F, \text{ where } M_w = M_{w_1} \dots M_{w_n} \text{ for } w = w_1 \dots w_n.$$

- LRS are weighted automata for $|\Sigma| = 1$.
- We consider extensions of LRS from this point of view.

We work over $\mathbb{Q}$

Polynomial sequences (polyrec)

Change matrices M_a to polynomial maps

Polynomial automata, cost-register automata, polynomial recurrent relations

[Benedikt et al., 2017], [Alur et al., 2013], [Sénizergues, 2007]

Example $a_n = 2^{2^n}$

$$a_0 = 2, \quad a_{n+1} = a_n^2, \quad f_1(x) = x^2$$

Example $a_n = n!$

$$a_0 = 1, \quad a_{n+1} = a_n \cdot b_n, \quad f_1(x, y) = xy$$

$$b_0 = 1, \quad b_{n+1} = b_n + 1, \quad f_2(x, y) = y + 1$$

Example $a_n = 2^{2^n} + n!$, or $a_n = 2^{2^n} \cdot n!$

Holonomic/P-recursive sequences

In the automata definition generalise to context-free grammars (CFGs)

Weighted/probabilistic CFGs [Chomsky and Schützenberger, 1959]

Equivalently: allow polynomials (in n) in the recursion

Example $a_n = n!$

$$a_0 = 1, \quad a_{n+1} = a_n \cdot n$$

Example C_n Catalan numbers

$$C_0 = 1, \quad (n+2)C_{n+1} = (4n+2)C_n$$

Example $a_n = C_n + n!, \quad a_n = C_n \cdot n!, \quad a_n = C_n * n!$

MSO sequences

Present in the logic framework, replace quantifiers with aggregators

Weighted MSO [Droste and Gastin, 2005]

MSO formula evaluate to 0 or 1, constants evaluate to themselves,

$\wedge, \vee$ become $\cdot$ and $+$, $\exists, \forall$ become $\sum$ and $\prod$

Example $a_n = n^n$ and $b_n = n!$

$$a_n = \prod_x \sum_y 1, \quad b_n = \prod_x \sum_y [x \leq y]$$

Example $a_n = 2^n$ and $b_n = 2^{2^n}$

$$a_n = \sum_X 1, \quad b_n = \prod_X 1$$

Three nonlinear extensions

Asymptotics of sequences

Bound on $|a_n| \leq f(n)$?

Conclusion

- Boundedness proof is inspired by [Simon 1994]

we believe it should work in more dimensions